THE UNIVERSITY OF SYDNEY

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CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Mathematics and Statistics

EXAMINATION

Semester 1 - Final, 2023

MATH2921 Vector Calculus and Differential Eqs (Adv) (Paper)

EXAM WRITING TIME: 2 hours

READING TIME: 10 minutes

EXAM CONDITIONS:
1. Closed book: no reference materials/resources are permitted

MATERIALS PERMITTED IN THE EXAM VENUE:
(No electronic aids are permitted e.g. laptops, phones)

Blank scratch paper - multiple sheets

Formula sheet (provided in the quiz/exam paper by unit coordinator)

MATERIALS TO BE SUPPLIED TO STUDENTS:
1 x 8-page answer book

INSTRUCTIONS TO STUDENTS:

Please tick the box to confirm that your examination paper is complete. ☐

For Examiner Use Only

Q	Mark
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Total _____

Multiple Choice Section

There is one and only one correct choice for each question. Please circle the right answer. There are 15 MCQs in all.

Question 1 **Points: 2**

For (x, y) and (r, θ) being the Cartesian and polar coordinates for $\mathbb{R}^2$, which of the following is true for a $2D$ region $\mathcal{R}$?

(A) $\iint_{\mathcal{R}} \frac{1}{\sqrt{x^2+y^2}} dxdy = \iint_{\mathcal{R}} 1 drd\theta$

(B) $\iint_{\mathcal{R}} 1 dxdy = \iint_{\mathcal{R}} 1 drd\theta$

(C) $\iint_{\mathcal{R}} \sqrt{x^2+y^2} dxdy = \iint_{\mathcal{R}} 1 drd\theta$

(D) $\iint_{\mathcal{R}} \sqrt{x^2+y^2} dxdy = \iint_{\mathcal{R}} r drd\theta$

Question 2 **Points: 2**

For the sphere $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$, the vector field $\vec{n}(x, y, z) = (x, y, z)$ and the ball $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$, which of the following is true?

(A) $\iiint_B (x + y + z) dV = \iint_S (x, y, z) \cdot \vec{n} dS$

(B) $\iint_S (x, y, z) \cdot \vec{n} dS = 0$

(C) $\iiint_B y^2 dV = 0$

(D) $\iint_S (z, y - x^2, zx + y) \cdot \vec{n} dS = \iiint_B (x + 1) dV$

Question 3 **Points: 2**

Consider a smooth function $f(x, y)$ over the region $\mathcal{R} = \{(x, y) \mid 0 < x < 1, 0 < y < 1\}$, which of the following is true?

(A) $f(x, y)$ can always be extended continuously to $\overline{\mathcal{R}} = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$.

(B) $f(x, y)$ might not have any critical point in $\mathcal{R}$.

(C) $f(x, y)$ always attains maximum and minimum in $\mathcal{R}$.

(D) $f(x, y)$ has at least 1 critical point in $\mathcal{R}$.

Question 4 **Points: 2**

For a $3D$ region V in $\mathbb{R}^3$ which contains a unit ball, which of the following must be true for the triple integral $\iiint_V 1 dV = \iiint_V 1 \, dxdydz$?

(A) It is equal to $\iiint_V r \sin\theta drd\theta dz$ in cylindrical coordinates.

(B) It can be negative.

(C) None of the other three.

(D) It is no bigger than $\iiint_V \rho^2 d\rho d\theta d\varphi$ in spherical coordinates.

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Question 5 **Points: 2**

For the function $f(x,y) = \det \begin{bmatrix} A(x) & B(y) \\ C(x) & D(y) \end{bmatrix} = \begin{vmatrix} A(x) & B(y) \\ C(x) & D(y) \end{vmatrix}$, which of the following is equal to $f_x(x,y)$?

(A) $\begin{vmatrix} A'(x) & B(y) \\ C(x) & D(y) \end{vmatrix} + \begin{vmatrix} A(x) & B(y) \\ C'(x) & D(y) \end{vmatrix}$

(B) $\begin{vmatrix} A'(x) & 0 \\ C'(x) & 0 \end{vmatrix} = 0$

(C) $\begin{vmatrix} A'(x) & B(y) \\ C'(x) & D(y) \end{vmatrix}$

(D) $\begin{vmatrix} A'(x) & B(y) \\ C(x) & 0 \end{vmatrix} + \begin{vmatrix} A(x) & 0 \\ C'(x) & D(y) \end{vmatrix}$

Question 6 **Points: 2**

Let $f(x)$ be a continuous function on $[-\pi, \pi]$ with the Fourier series $F(x)$ of period 2π. Which of the following is true?

(A) $F(-\pi)$ might not be equal to $F(\pi)$.

(B) $F(\pi) = f(\pi)$ since $f(x)$ is continuous up to π.

(C) $F(-\pi)$ might not be equal to $f(-\pi)$.

(D) $f(x)$ can always be extended to a continuous function of period 2π over $x \in \mathbb{R}$.

Question 7 **Points: 2**

In $\mathbb{R}^2$, C is the unit circle centred at the origin in the counter-clockwise direction. Which of the following is true?

(A) $\int_C ds = \pi$

(B) $\int_{-C} (y,x) \cdot d\vec{r} = \int_C (-y,-x) \cdot d\vec{r}$

(C) $\int_{-C} ds = -2\pi$

(D) $\int_C (x,y) \cdot d\vec{r} > 0$

Question 8 **Points: 2**

Which of the following is a conservative vector field?

(A) $(y, 0)$

(B) $(-y, x)$

(C) $(y, 2x)$

(D) $(0, y)$

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Question 9 **Points: 2**

Which of the following is always a normal vector for the level surface $\{(x, y, z) \mid z + f(x, y) = 0\}$ in $\mathbb{R}^3$ at the point $(0, 0, -f(0, 0))$?

(A) $(-f_y(0, 0), f_x(0, 0), 1)$

(B) $(f_x(0, 0), f_y(0, 0), -1)$

(C) $(f_x(0, 0), f_y(0, 0), 1)$

(D) $(f_x(0, 0), f_y(0, 0), 0)$

Question 10 **Points: 2**

Which pair of functions would form a fundamental set of solutions for the ODE with unknown $y(x)$,

$$y'' - 3y' + 2y = 0.$$

(A) $x, 2x$

(B) e^{-x}, e^{-2x}

(C) $\sin(x), \sin(2x)$

(D) e^{x}, e^{2x}

Question 11 **Points: 2**

Let $A(x)$ be a 2×2 matrix with entries being functions with the variable x. Which of the following is true?

(A) $e^{A(x)}$ is always invertible for any x.

(B) $e^{A(x)}$ might have determinant 0 for some x.

(C) $A(x)e^{A(x)} = e^{A(x)}A(x)$ might not hold for some x.

(D) $(e^{A(x)})' = A'(x)e^{A(x)}$ is always true.

Question 12 **Points: 2**

Which of the following should be the guess for a particular solution for the ODE with unknown $y(x)$,

$$y'' + 4y' + 4y = x^2e^x$$

(A) $(Ax^2 + Bx + C)e^x$

(B) Ax^2e^x

(C) $(Ax^2 + Bx + C)e^{-2x}$

(D) Axe^{-2x}

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Question 13 **Points: 2**

Let y_1 and y_2 be two solutions of the ODE with unknown $y(x)$,

$$y'' + 3y' + 2y = \cos(x) + \sin(x).$$

Which of the following is true?

(A) $y_1 - y_2$ solves $y'' + 3y' + 2y = \cos(x) + \sin(x)$.

(B) $y_1 + y_2$ solves $y'' + 3y' + 2y = \cos(x) + \sin(x)$.

(C) $y_1 - y_2$ solves $y'' + 3y' + 2y = 0$.

(D) It is possible that $y_1 = 2y_2$.

Question 14 **Points: 2**

Which of the following is true about the type of the ODE with unknown $y(x)$,

$$y' - y^2 = 4?$$

(A) first order separable ODE

(B) linear first order ODE

(C) second order separable ODE

(D) nonlinear second order ODE

Question 15 **Points: 2**

Which of the following is true about the Wronskian $W(x)$ of a fundamental set of solutions for the first order linear homogeneous ODE system

$$\vec{y}\,'(x) = A(x)\vec{y}(x)?$$

(A) $W(x)$ is identically 0, in order words, the constant function of value 0.

(B) $W(x)$ must be equal to 0 for some x.

(C) $W(x)$ can't be equal to 0 for any x.

(D) None of the other three is true.

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Extended Answer Questions

Marks will be awarded for working. There are 3 questions in all, all with multiple parts. Show the working in the space provided below each question.

Question 16 **Points: 13**

This question is about differential equations.

(a) For the unknown $u(x,t)$ over $\mathbb{R} \times \mathbb{R}$, solve

$$u_{tt} - 4u_{xx} = 0, \quad u(x,0) = \sin x, \quad u_t(x,0) = \cos(2x).$$

(b) For the unknown $u(x,y)$ over $[0,\pi] \times [0,2\pi]$, solve

$$u_{xx} + u_{yy} = 0, \quad u(x,0) = u_x(0,y) = u_x(\pi,y) = 0, \quad u(x,2\pi) = -\cos(2x).$$

(c) For the unknown $y(x)$, solve

$$y'' - 2y' + y = e^x, \quad y(0) = 0, \quad y(1) = 1.$$

Show your working in the space provided below.

Question 17 **Points: 9**

This question is about integration.

(a) Calculate the following integrals.

(a.1) For $V = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 9\}$ in Cartesian coordinates,

$$\iiint_V \frac{\sin(x^2 + y^2 + z^2)}{\sqrt{x^2 + y^2 + z^2}}\, dxdydz$$

(a.2) For $R = \{(x, y) \mid x \geq 0, y \geq 0, x + y \leq 1\}$ in Cartesian coordinates,

$$\iint_R x\, dxdy.$$

(b) Consider $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 25,\ 0 \leq z \leq 4\}$ and the vector field $\overrightarrow{F}(x, y, z) = (y, -z, x)$. Applying Stokes' Theorem or not, calculate

$$\iint_S (\nabla \times \overrightarrow{F}) \cdot \overrightarrow{n}\, dS$$

where $\overrightarrow{n}$ is the unit normal vector field of S pointing in the direction of leaving the origin.

Show your working in the space provided below.

Question 18 **Points: 8**

This question is comprehensive.

(a) Let $H(x, y) = 2x^2 - 2xy + 2y^2$. Consider the vector field

$$\vec{F}(x, y) = (H_y, H_x).$$

The curve $\vec{r}(t) = (x(t), y(t))$ satisfies

$$\vec{r}\,'(t) = \vec{F}(\vec{r}(t)).$$

(a.1) Write down the 1st order ODE system for the unknown $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ from $\vec{r}(t)$.

(a.2) Solve the first order ODE system from (a.1), then get the curve $\vec{r}(t)$ starting from $(2, 1)$, in other words, $\vec{r}(0) = (2, 1)$.

(b) $F(t)$ is a smooth function of variable t. Consider the following integral equation of unknown $y(x)$,

$$y = \int_0^y F(t)dt + \sin x.$$

(b.1) Rewrite the integral equation as a first order ODE of unknown $y(x)$ by taking derivative.

(b.2) If $F(0) = -1$, for the curve of the particular solution of the ODE from $(b.1)$ satisfying $y(0) = 0$, find its tangent line at the origin in the xy-plane $\mathbb{R}^2$.

(c) In $\mathbb{R}^2$, consider a conservative vector field $\vec{F}(x, y) = (f(x, y), g(x, y))$. We define its potential $\Phi(x, y)$ as follows. At the point (a, b), we set $\Phi(a, b) = 0$. Then for any other point (x, y), we pick a curve C from (a, b) to (x, y), and set $\Phi(x, y) = \int_C \vec{F} \cdot d\vec{r}$ which is independent of the choice of such curve C because $\vec{F}$ is conservative. Show $\Phi_x(a, b) = f(a, b)$ and $\Phi_y(a, b) = g(a, b)$ through the following steps.

(c.1) For the point $(a + s, b)$, consider the curve C_s, $\vec{r}(t) = (a + ts, b)$ for $t \in [0, 1]$. Calculate $\Phi(a + s, b) = \int_{C_s} \vec{F} \cdot d\vec{r}$.

(c.2) Using the result from (c.1), calculate $\Phi_x(a, b)$ as $\frac{d\Phi(a+s,b)}{ds}$ at $s = 0$ to see it's equal to $f(a, b)$.

(c.3) Similarly, show $\Phi_y(a, b) = g(a, b)$.

Show your working in the space provided below.

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Formula Sheet

Formulas and theorems provided might be stated without the complete conditions under which they apply. The notations used are the same as those used in lectures.

Line Integrals

$$\int_{\mathcal{C}} \phi(x,y,z)\, ds = \int_a^b \phi(x(t),y(t),z(t))\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

$$\int_{\mathcal{C}} \overrightarrow{F} \cdot d\overrightarrow{r} = \int_{\mathcal{C}} F_1\, dx + F_2\, dy + F_3\, dz = \text{Work done by } \overrightarrow{F} \text{ along } \mathcal{C}$$

Gradient (in $\mathbb{R}^3$ for example)

$$\text{grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\overrightarrow{i} + \frac{\partial\phi}{\partial y}\overrightarrow{j} + \frac{\partial\phi}{\partial z}\overrightarrow{k}$$

If $\overrightarrow{F} = \nabla\phi$ for a smooth function ϕ, then equivalently $\overrightarrow{F}$ is a conservative vector field, ϕ is a potential function of $\overrightarrow{F}$, and $\int_{\mathcal{C}} \overrightarrow{F} \cdot d\overrightarrow{r}$ is path-independent.

Curl (for $\overrightarrow{F} = (F_1, F_2, F_3)$)

$$\text{curl } \overrightarrow{F} = \nabla \times \overrightarrow{F} = \begin{vmatrix} \overrightarrow{i} & \overrightarrow{j} & \overrightarrow{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

If the domain of $\overrightarrow{F}$ has no hole and $\nabla \times \overrightarrow{F} = \overrightarrow{0}$ then $\overrightarrow{F}$ is conservative.

Divergence: $\text{div}\, \overrightarrow{F} = \nabla \cdot \overrightarrow{F}$

Double Integral

$$Area(\mathcal{R}) = \iint_{\mathcal{R}} dxdy$$

Signed volume under the graph of $z = f(x,y)$ over $\mathcal{R}$ is

$$\iint_{\mathcal{R}} f(x,y)\, dxdy$$

In polar coordinates,

$$\iint_{\mathcal{R}} f(x,y)\, dxdy = \iint_{\mathcal{R}} f(r\cos\theta, r\sin\theta)\; rdrd\theta$$

Triple Integral

In *Cartesian coordinates* (x, y, z):

$$\iiint_V f(x,y,z)\,dV = \iiint_V f(x,y,z)\,dxdydz.$$

In *cylindrical coordinates* (r, θ, z):

$$\iiint_V f(x,y,z)dV = \iiint_V f(r\cos\theta, r\sin\theta, t)\,rdrd\theta dz.$$

In *spherical coordinates* (ρ, θ, φ):

$$\iiint_V f(x,y,z)\,dV = \iiint_V f(\rho\cos\theta\sin\varphi, \rho\sin\theta\sin\varphi, \rho\cos\varphi)\,\rho^2\sin\varphi d\rho d\theta d\varphi$$

Green's Theorem/Formula (in $\mathbb{R}^2$)

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot d\overrightarrow{r} = \oint_{\mathcal{C}} F_1\,dx + F_2\,dy = \iint_{\mathcal{R}} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dxdy$$

with $\mathcal{C}$ bounding $\mathcal{R}$ in the counter-clockwise way. Green's Theorem is equivalent to **Divergence Theorem in 2D**

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot\overrightarrow{n}\,ds = \iint_{\mathcal{R}} \nabla\cdot\overrightarrow{F}\,dxdy$$

Surface Integral (for the graph S of $z = f(x, y)$ over $\mathcal{R}$)

$$\iint_S \phi(x,y,z)\,dS = \iint_{\mathcal{R}} \phi(x,y,f(x,y))\sqrt{1+f_x^2+f_y^2}\,dxdy$$

In a general coordinate $\overrightarrow{T}(u,v)$, $\sqrt{1+f_x^2+f_y^2}dxdy$ is replaced by $\|\overrightarrow{T}_u \times \overrightarrow{T}_v\|dudv$.

Flux of $\overrightarrow{F}$ through S in the direction of (unit normal) $\overrightarrow{n}$ is

$$\iint_S \overrightarrow{F}\cdot\overrightarrow{n}\,dS.$$

Gauss' Theorem (Divergence Theorem in 3D)

$$\iint_S \overrightarrow{F}\cdot\overrightarrow{n}\,dS = \iiint_V \nabla\cdot\overrightarrow{F}\,dV$$

for the unit outward normal $\overrightarrow{n}$ of the surface S which the boundary of the $3D$ region V.

Stokes' Theorem

$$\oint_{\mathcal{C}} \overrightarrow{F}\cdot d\overrightarrow{r} = \iint_S (\nabla\times\overrightarrow{F})\cdot\overrightarrow{n}\,dS$$

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where $\overrightarrow{n}$, $\mathcal{C}$ and S are compatible.

Abel's Identity/Formula

For the ODE over $[a, b]$,

$$Ly = y'' + p(x)y' + q(x)y = 0,$$

the Wronskian of 2 solutions $W(y_1, y_2)$ satisfy that for any $x_0 \in [a, b]$,

$$W' = -p(x)W, \quad W(x) = W(x_0)e^{-\int_{x_0}^{x} p(t)dt}.$$

For the $n \times n$ ODE system over $[a, b]$,

$$\overrightarrow{y}'(x) = A(x)\overrightarrow{y}(x),$$

the Wronskian of n solutions $W(\overrightarrow{y}_1, \cdots, \overrightarrow{y}_n)$ satisfies for any $x_0 \in [a, b]$,

$$W' = \mathrm{Tr}\big(A(x)\big)W, \quad W(x) = W(x_0)e^{\int_{x_0}^{x} \mathrm{Tr}\left(A(t)\right)dt}.$$

Variation of Constants

For the first order linear ODE system

$$\overrightarrow{y}'(x) = A(x)\overrightarrow{y}(x) + \overrightarrow{f}(x),$$

given the Fundamental Solution Matrix $\Phi(x)$ for the corresponding homogeneous system, the general solution is

$$\overrightarrow{y}(x) = \Phi(x)\int_{x_0}^{x} \Phi^{-1}(t)\overrightarrow{f}(t)dt + \Phi(x)\overrightarrow{C}$$

for any constant vector $\overrightarrow{C}$.

Variation of Parameters

For the second order linear ODE,

$$y'' + p(x)y' + q(x)y = f(x),$$

given the fundamental set of the corresponding homogeneous ("complementary") ODE $\{y_1, y_2\}$, the general solution is

$$y = u_1(x)y_1(x) + u_2(x)y_2(x), \quad u_1 = -\int \frac{y_2 \cdot h}{W(y_1, y_2)}\, dx, \quad u_2 = \int \frac{y_1 \cdot h}{W(y_1, y_2)}\, dx.$$

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Fourier Series

For a function f defined over $[-L, L]$ for some $L > 0$, the *Fourier series* is:

$$\mathcal{F}(f)(x) = a_0 + \sum_{n=1}^{\infty}[a_n \cos\left(\tfrac{n\pi}{L}x\right) + b_n \sin\left(\tfrac{n\pi}{L}x\right)]$$

where the *Fourier coefficients* b_0, $(b_n)_{n\geq 1}$ and $(a_n)_{n\geq 1}$ are given by

$$\begin{aligned} a_0 &= \tfrac{1}{2L}\int_{-L}^{L} f(x)\,dx \\ a_n &= \tfrac{1}{L}\int_{-L}^{L} f(x)\cos\left(\tfrac{n\pi}{L}x\right)\,dx \qquad \text{for every } n \geq 1, \\ b_n &= \tfrac{1}{L}\int_{-L}^{L} f(x)\sin\left(\tfrac{n\pi}{L}x\right)\,dx \qquad \text{for every } n \geq 1. \end{aligned}$$

For $y'' + \lambda y = 0$, we have the following **eigenvalue problems**:

- Eigenvalue Problem (a) on $[0, L]$: $y(0) = y(L) = 0$; eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunction $\sin(\frac{k\pi}{L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (b) on $[0, L]$: $y'(0) = y'(L) = 0$; eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunction $\cos(\frac{k\pi}{L}x)$ for $k = 0, 1, 2, \cdots$
- Eigenvalue Problem (c) on $[0, L]$: $y'(0) = y(L) = 0$; eigenvalue $\lambda = \frac{(2k-1)^2\pi^2}{4L^2}$ with eigenfunction $\cos(\frac{(2k-1)\pi}{2L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (d) on $[0, L]$: $y(0) = y'(L) = 0$; eigenvalue $\lambda = \frac{(2k-1)^2\pi^2}{4L^2}$ with eigenfunction $\sin(\frac{(2k-1)\pi}{2L}x)$ for $k = 1, 2, \cdots$
- Eigenvalue Problem (e) on $[-L, L]$: $y(-L) = y(L) = 0, y'(-L) = y'(L)$; eigenvalue 0 for eigenfunction 1, eigenvalue $\lambda = \frac{k^2\pi^2}{L^2}$ with eigenfunctions $\cos(\frac{k\pi}{L}x)$ and $\sin(\frac{k\pi}{L}x)$ for $k = 1, 2, \cdots$

For $u_{tt} - C^2 u_{xx} = 0$ with constant C for $(x, y) \in \mathbb{R}^2$ and the initial conditions, $u(x, 0) = f(x), u_t(x, 0) = g(x)$, the D'Alembert's Solution is

$$u(x,t) = \frac{1}{2}[f(x + Ct) + f(x - Ct)] + \frac{1}{2C}\int_{x-Ct}^{x+Ct} g(s)ds.$$

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Table of Standard Integrals

1. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

2. $\int \frac{dx}{x} = \ln|x| + C$

3. $\int e^x\,dx = e^x + C$

4. $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$

5. $\int \sin x\,dx = -\cos x + C$

6. $\int \cos x\,dx = \sin x + C$

7. $\int \sec^2 x\,dx = \tan x + C$

8. $\int \sinh x\,dx = \cosh x + C$

9. $\int \cosh x\,dx = \sinh x + C$

10. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$

11. $\int x\,\sin x\,dx = -x\cos x + \sin x + C$

12. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

13. $\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1}\frac{x}{a} + C = \ln\left(x + \sqrt{x^2+a^2}\right) + C'$

14. $$\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1}\frac{x}{a} + C \quad (x > a)$$
$$= \ln\left|x + \sqrt{x^2-a^2}\right| + C' \quad (x > a \text{ or } x < -a)$$

This is the end of the exam paper.